

ApparelPro

Enterprise Resource Planning for
Garment & Apparel Manufacturing

Product Overview & Capabilities

12+

Core Modules

98

Database Tables

146

API Endpoints

2,752

Source Files

September 2026

Executive Summary

ApparelPro is a comprehensive Enterprise Resource Planning (ERP) system purpose-built for the garment and apparel manufacturing industry. It streamlines the entire production lifecycle — from purchase order intake and style configuration through fabric procurement, production scheduling, quality control, and commercial invoicing for export shipments.

Built on a modern three-tier architecture with React 19 for the user interface, .NET 10 Web API for business logic, and SQL Server 2022 for data management, ApparelPro delivers a responsive, secure, and scalable solution that can be deployed on-premise or in the cloud using Docker containers orchestrated by Coolify with Cloudflare providing global edge security.

Application Modules

Order Management

End-to-end purchase order processing and tracking

- Buyer registration and multi-currency management
- Purchase order creation with garment type classification
- Season, basis (FOB/CMP/CMT), and destination tracking
- Order quantity overrides with audit trail

Style Management

Complete style master configuration for each order

- Style creation with composite key (Buyer/Order/Type/Style)
- Color-size ratio and quantity matrix management
- Unit pricing and export balance tracking
- Style cloning across orders

Material Management

Fabric, trim, and accessory procurement and consumption

- Bill of Materials (BOM) with material cost profiling
- Material consumption ledger tracking issued vs consumed
- Supplier purchase order generation from material requirements
- Multi-unit conversion (yards, meters, dozens, pieces)

Production Planning

Production line scheduling and daily tracking

- Production line allocation with capacity planning
- Daily production entries with time ticket tracking
- Estimated vs actual production comparison
- Operation breakdown with SAM (Standard Allowed Minutes)

Inventory & Stock

Order-wise and general warehouse management

- Order-wise stock tracking (received, issued, returned, damaged)
- General store management for non-order materials
- Stock transaction ledger with full audit trail
- Inter-department material transfers and requisitions

Commercial & Export

Export documentation and commercial invoicing

- Commercial invoice generation with documentary buyer details
- Part-shipment tracking against order quantities
- LC (Letter of Credit) and assessment number management
- Carrier and port-of-loading configuration

Import / Export Documentation

Comprehensive customs and shipping document management

- Certificate of Origin, Customs Declaration, and Export License generation
- Packing lists with carton details and string-up specifications
- Boat notes and value declarations for shipment processing
- Reference codes: clearance offices, payment terms, transport modes, duty/tax codes

Subcontracting

Outsourced process management

- Subcontractor registration and capability tracking
- Subcontract creation linked to specific styles
- Additional processing cost allocation
- Subcontract completion and quality sign-off

Reporting & Analytics

Comprehensive business intelligence

- 155 built-in report templates across all modules
- Production efficiency dashboards
- Material cost analysis and variance reports
- Export shipment and order pipeline status

Event & Pipeline Management

Order lifecycle tracking

- Configurable event milestones per style
- Order pipeline stage tracking with date history
- Estimated production end-date management
- Approval workflow for estimates

Reference Data

Master data configuration

- Buyer, supplier, and subcontractor directories
- Item feature and garment component catalogs
- Unit of measure and conversion tables
- Department, section, and employee registers

User & Security

Authentication and authorization

- JWT-based token authentication
- Role-based access control (Merchandiser, Manager, Operator, Inventory)
- Microsoft Identity integration
- Audit trail on all critical transactions

Document Management

Automated document sequencing

- Auto-incrementing document numbers by type
- Prefix-based document identification
- Transaction-safe sequence allocation
- Receipt, PO, and invoice numbering

Why ApparelPro

Industry-Specific

Purpose-built for garment manufacturing with deep domain understanding of style configurations, color-size matrices, fabric consumption, and export documentation workflows.

Modern Architecture

Three-tier architecture with React 19 frontend, .NET 10 API, and SQL Server 2022 ensures performance, maintainability, and access to the latest framework capabilities.

Scalable Deployment

Docker-containerized deployment with Coolify orchestration and Cloudflare edge security. Supports on-premise, cloud, or hybrid deployment models.

Comprehensive Coverage

12+ integrated modules with 98 database tables covering the entire garment manufacturing lifecycle from order intake to export shipment.

Role-Based Security

Fine-grained access control with JWT authentication ensuring each user sees only what their role permits, with full audit trail on critical operations.

Real-Time Analytics

155 built-in reports provide instant visibility into production efficiency, material costs, order pipeline status, and financial performance.